Change-Point Detection of Time-series Data in the Blogosphere

Kazuya Satou[1], Hideki Takayasu[2], and Misako Takayasu[3]

[1,3] Tokyo Institute of Technology 4259,Nagatsuta-cho, Midori-ku, Yokohama-shi, Kanagawa, 226-8502 Japan

[2] Sony Computer Science Laboratories, 3-14-13, Higashigotanda, Shinagawa-ku, Tokyo, 141-0022 Japan

[2] Meiji Institute for Advanced Study of Mathematical Sciences, 4-21-1 Nakano, Nakano-ku, Tokyo, 164-8525 Japan

E-mail: [1]satou@smp.dis.titech.ac.jp, [2]takayasu@csl.sony.co.jp, [3]takayasu@smp.dis.titech.ac.jp

Abstract

Time series of the words written in the blogs can be classified into four types by the frequency of occurrence: daily, boom, seasonal, and news words. In this paper, we introduce a general method of change-point detection for complicated time series, and apply it for segmentation and characterization of blog time series.

Keyword: Change-point Detection, Blogosphere

1. Introduction

There are many methods to detect change points of the time series [1,2]. However, most of these methods are parametric, and are not suitable for general time series. We propose a nonparametric method based on Fisher's exact test [3,4] to apply for blog time series.

2. Data

We analyze the Japanese blog data since November 2006 provided by Dentsu Buzz Research (about 1.8 billion article). In order to eliminate the periodicity, we normalize the number of blogs containing a certain keyword per day by the total number of blogs [5]. This value $r^{(k)}(t)$ is the frequency of occurrence of keyword k, and we analyze the time series of $r^{(k)}(t)$.

3. Method

First, for given window of time series $t \in [t_L, t_R]$, we set thresholds, the time t_{th} and the frequency of occurrence r_{th}, and define the contingency table in order to apply the Fisher's exact test to time series.

In the Table1, the p-value is given by Eq.(1) from combinatorics.

Table1.the contingency table for the time series

	$t_L \le t \le t_{th}$	$t_{th} < t \le t_R$	Total
$r^{(k)}(t) > r_{th}$	a	b	a + b
$r^{(k)}(t) \le r_{th}$	c	d	c + d
Total	a + c	b + d	n

$$p = \frac{(a+b)!\,(c+d)!\,(a+c)!\,(b+d)!}{n!\,a!\,b!\,c!\,d!} \quad (1)$$

Second, to detect trends of various periods, we introduce moving windows of different size $n, \{n \mid 14x, x = 2,3,\ldots,10\}$.

Finally, we calculate the p-value for 100 thousand randomized time series, and define the 100th p-value in ascending order as the significance level.

4. Result

We show the result for "KY"[1] in Figure1. The black line is the time series of the frequency of the occurrence, and the background color shows the result. If the p-value in a certain date is in a significant level, we plot the colored line. And the colors are red (ascending) and blue (descending) determined by the sign of $D = (bc - ad)$. The result is consistent with our intuition.

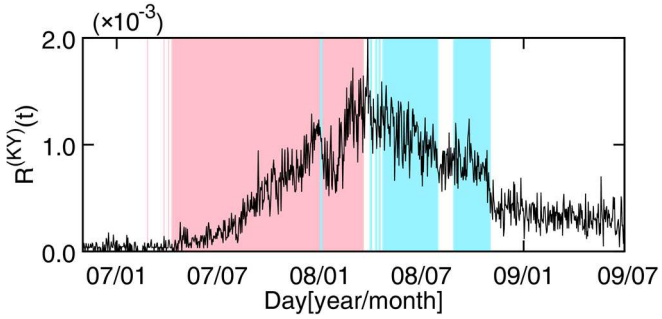


Figure1: The time series and the detection result

5. Conclusion

Since the numerical result of our method is consistent with our intuition, an automated change-point detection is expected. We will compare the results of our method with the ones of the other methods. We will study with a view to the implementation such as the search system and the alerting.

References

[1] J.Takeuchi, K.Yamanishi, IEEE Transactions On Knowledge And Data Engineering, 18, 4 (2006)

[2] G.Chen, Y.K.Choi, Y.Zhou, Jounal Of Econometrics, 143, 2, 227-262 (2008)

[3] R.A.Fisher, J.R.STAT.SOC, 85, 1, 87-94 (1922)

[4] A.Sato, H.Takayasu, arXiv:1309.0602 (2013)

[5] Y.Sano, K.Yamada, H.Watanabe, H.Takayasu, M.Takayasu, Phys.Rev.E, 87, 012805 (2013)

[1] Japanese abbreviated word of "Kuuki Yomenai", meaning that "cannot read the atmosphere"